Avinash Tingre

Senior Software Engineer | Payments Platforms | Java, Spring Boot, Distributed Systems
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Summary

Senior Software Engineer with 9 years of experience building payment platforms in Java and Spring Boot, the last 4+ on Mastercard Checkout Solutions (Click to Pay). Currently the design owner and epic lead for a 40M+ card data migration, working day to day with engineering, DBA, SRE and architecture. Halved p90 latency on core directory APIs, raised unit test coverage from 75% to 90%, moved 7 repositories to cleaner branching and merge practices, and onboarded 3 engineers.

Experience

- **Mastercard** Pune, India
  Senior Software Engineer *Feb 2024 - Present*

  Card Management Service, Mastercard Checkout Solutions *Sep 2025 - Present*
  - Design owner and epic lead for migrating 40M+ card records from the legacy issuer directory to the Directory 2.0 platform that powers P2P payments. Wrote both low-level designs, ran design reviews, and coordinated 4-5 engineers, 2 architects, DBA and SRE through production DDL and cutover.
  - Halved p90 latency of the Directory 2.0 update API (40 ms to 20 ms) and cut the lookup API from 16 ms to 9 ms by running decryption in parallel, dropping unused token decryption, replacing N+1 queries with batched reads and capping async fan-out. Verified with Dynatrace and load tests.
  - Designed and shipped a paginated bulk id-lookup API that returns ids in a stable order with a continuation token (1,000 per page by default, 5,000 max, optional issuer filters), so migration and consumer jobs can resume where they left off instead of scanning full records.
  - Built the P2P enrollment tokenization flow and a standalone Tokenize API (MDES tokenization, encryption-service integration, rollback when a save fails) with full Karate coverage, then ran 4 regression cycles across 2 releases with no regressions.
  - Resolved a P2 production crash where about 1.5M verification records for a single card drove the JVM out of memory: found the root cause, coordinated the data cleanup and fixed the unbounded inserts. Also sized a 2.9M-row buildup of stale request metadata and wrote the cleanup plan.
  - Improved engineering practices across 7 repositories: unit test coverage from 75% to 90%, squash merges, cleanup of 720 of 1,013 stale branches (71%), primary code reviewer for 3 repos, onboarded 3 engineers, 20+ design and analysis pages. Spot Award and Game Changer badge.
  - Built a verification utility that samples migrated records, diffs them column by column across both directories and reports mismatches; defined the team's post-migration validation approach.
  - Built a config-driven batch framework (partitioning, SELECT FOR UPDATE row locks, chunked parallel processing, job tracking) to migrate millions of PAN hashes to HMAC for PCI DSS v4.0.
  - Completed the V1 to V2 card data-model migration (fixed the root causes of legacy fallbacks, disabled legacy writes, enabled V2 schedulers) and monitored the staged 5/25/100% production cutover.

  Authentication Orchestration, Mastercard Checkout Solutions *Feb 2024 - Sep 2025*
  - Delivered FIDO/passkey and MFA capabilities for Click to Pay authentication: eligibility checks by country, issuer, BIN range and MFA enablement, native-app integration checks across six B2B and B2C APIs, session state handling and assertion token generation.
  - Added alternate-site retry with circuit breaking for tokenization-service calls (active-active failover) and replaced auth-type-specific session endpoints with a single API. Move Fast achievement.
  - Fixed a Priority-1 BindMFA 503 failure in a shared test facility, built negative-scenario simulators for crypto validation and enriched the KPI events behind the BizOps liability-shift dashboard.

- **Opus Technologies (client: Mastercard, Click to Pay)** Pune, India
  Senior Software Engineer *Feb 2022 - Feb 2024*
  - Enhanced the Click to Pay checkout service (Java, Spring Boot, Redis) with new microservice integrations and checkout-path caching; shipped critical fixes in tight release windows. 3x Spot Awards.
  - Built a WireMock extension that finds unused stubs in functional test suites; removed 5,000 dead stubs and made the suite far easier to maintain.

- Designed an audit logging framework (Spring Boot, AOP) that records every API request across services for security review.
- Built the Checkout Toolkit web UI (ReactJS, Spring Boot), which internal and partner teams adopted for day-to-day testing.

- **Cerence** Pune, India
  Software Engineer *May 2021 - Feb 2022*
  - Built features and integrations for the Natural Language Processing Server in Java, and wrote a JSON-to-spreadsheet extraction utility that sped up data analysis for the team.

- **Amdocs** Pune, India
  Software Developer *Oct 2017 - Feb 2021*
  - Built a Spring Boot and Hibernate web service that moved data between in-house inventory and DNS management applications.
  - Automated Linux deployments with Ansible and shell scripts, cutting deployment time by 50% and errors by 30%. Spot Award.
  - Built internal tools used across the organization: a Hotfix Search app (Java, ReactJS, Apache POI) and a Leave Management System with auto-approval.

Skills

- **Languages:** Java 8 to 17, SQL, JavaScript, Shell, ReactJS
- **Backend:** Spring Boot, Spring Cloud, Spring Security, Spring Batch, Hibernate/JPA, REST/OpenAPI
- **Data:** Oracle (DDL, indexing, partitioning, query tuning), PostgreSQL, Redis, Kafka
- **Platform and observability:** Cloud Foundry, Docker, Jenkins, SonarQube, Git, Dynatrace, Splunk
- **Testing:** JUnit 5, Mockito, Karate, PACT, WireMock, load and performance testing
- **Reliability and security:** Retry, circuit breaker, active-active failover, idempotency, PCI DSS, HMAC, JWT, 3DS, FIDO2/passkeys, tokenization

Education

- **Savitribai Phule Pune University** Pune, India
  Bachelor of Engineering, Computer Science *2017*